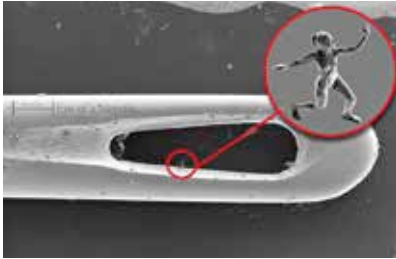


to unleash their creativity in multiple ways. Light, Darkness and the Tree by Se Yoon Park, Gyre by Charles-Eric Gogny and White Rabbit by Danny van Ryswyk are some exquisite sculptures that have been 3D printed. Jonty Hurwitz created, possibly, the smallest sculpture ever created with 3D printing technology. The micro-figurine, unfortunately, was lost, probably owing to its size.



Miniscule 3D printed sculpture that looks like a speck of dust from a distance

05 VEHICLES

3D printing is used in building parts of various vehicles such as cars, planes and trains. It can play a pivotal role in the future of manufacturing vehicles. It is also being used to build vehicles from scratch entirely with 3D printing technology. Rolls Royce uses 3D printing to build both, airplanes and cars. When it comes to trains, the technology is used only to build models. Recently, Formula 1 race cars also started using 3D printing, with certain teams bringing the printer to the racetrack for any last minute alterations in their cars.

06 LARGE-FORMAT CAMERAS

3D printing has been used to build pinhole cameras, with the image being captured using old school photosensitive film. The lenses make it tricky to 3D print more conventional cameras, such as a DSLR. A large-format pin hole camera has been 3D printed. One limitation of such cameras is the shutter mechanism since pinhole photography leaves the shutter open for too long and may result in image blur. This is because multiple impressions and movements are being recorded due to long exposure times.

07 CLOTHING

It may seem ridiculous to buy 3D printed clothing which is usually exorbitantly priced when compared to normal clothing that has been mass produced. However, high-end fashion brands are doing exactly this. They are 3D printing clothing with complex, geometric designs that are allegedly guaranteed to fit a particular body shape. People are buying 3D printed clothing since they look unique and cutting-edge. In the future, we could possibly browse an e-commerce website and 3D print a piece of clothing that we have purchased right at home.

3D printing is also utilised in making lightweight bulletproof clothing. Researchers used biomimicry of fish scales to develop a Flexible Scale Armour, which is an advanced bulletproof vest.



A store with 3D printed clothing

08 PARTS OF HOUSEHOLD APPLIANCES

Imagine having the ability to conjure up parts of household appliances that are broken or damaged in the comfort of your home. We could avoid situations like a plumber or technician visiting a house to fix an appliance, only to tell the homeowner that they need to come back once the spare part is available. Happy3D is a service by a French company, Boullanger, which allows their customers to download 3D printable files of missing or damaged parts of appliances and 3D print the replacement parts in their own homes or hubs.

09 DRONES

One of the greatest advantages of building objects and products with 3D printing is the ability to create them without

screws and nails which makes the object much lighter than traditionally-constructed objects. Southampton University put this idea into good effect by creating a drone which has no screws and nails at all. It has a 2-metre wingspan and can fly at speeds up to 90 miles per hour. This drone is almost silent because of its weight and only takes about ten minutes to assemble. Armies and navies have also begun testing 3D printed drones for various missions.

10 BABY MODELS

A Russian firm has created a tool that allows expecting parents to 3D print a model of their unborn child. This model made from ultrasound images that provide a 360-degree view of the baby in-utero. These images can be used to create 3D printed models that show the features of the baby in stunning detail. If you think this is creepy, think again, since many parents have used this tool and found physical abnormalities in their babies in advance which helped them be better prepared to raise them. Additionally, 3D printed models of babies are also used to perform simulated operations. Pediatric simulated operations were done with toy dolls in the past, however, the ability to print functioning baby organs in the models allow doctors to practice on them, which wasn't a possibility before.



Expecting mothers are 3D printing models of their unborn child

11 SHOES

3D printed shoes may be the solution to the woes of people with odd feet - large feet, uneven foot sizes or deformed shapes of feet. Standard shoes, many times, do not provide for the peculiar needs of people. 3D printed shoes are customised to match the purchaser's exact